



MINI-PCs

Powerful yet small

Small form-factor machines have especially benefited from the Industry moving to smaller manufacturing nodes for transistors. Processors have become faster while running cooler than before and graphics chips even more so. Graphics chips have improved to such an extent that both AMD and NVIDIA are offering the solutions for notebooks which are nearly equal to their desktop counterparts. The components race has greatly helped bring performance which was hitherto limited to bulky desktop machines into little cases that are barely larger than a school kid's lunchbox. Earlier in the year, Intel did make a few heads turn when they decided to shut the mobile SoC segment. Since mobile SoCs are quite similar to what was found in ultra small mini-PCs and stick PCs, it did raise a few questions about the survival of mini-PCs but when they ended up releasing NUCs with Skylake CPUs, all those worries were laid to rest. However, we are yet to see a rise in unit sales for these devices. There are high-end desktop units like the ASUS G20CB which nearly falls into the category that sport desktop Core i7s and GTX 1080s but even ASUS doesn't define them as mini-PCs. Instead they have the Vivo line up. For us, anything that occupies under five cubic litres is a mini-PC. So with that little parameter set in stone we can safely draw the line as to what is a mini-PC and what becomes a compact desktop. Given how these devices are not prevalent in India, we can conclude that there's something off about mini-PCs in India.

ZERO 1 WINNER

ZOTAC MAGNUS EN1060

Skylake socketed processor? Check. NVIDIA 1000 series graphics card? Check. Can it play Mine Craft? Double-check! While we did end up with fewer mini-PCs this year compared to last year, it wasn't going to be a surprise for a unit like the Magnus winning since it was the only one with a discrete graphics solution. So you have a desktop grade processor and a desktop grade graphics chip in a form-factor that doesn't even take up three cubic litres of space. Not only is this unit small enough to be discreetly hidden behind your TV, it's capable of running your current gen AAA titles for about 2-years without any worry.



ZOTAC MAGNUS EN1060

Price: ₹73,599



ASRock DeskMini H110

Price: ₹65,499



GIGABYTE BRIX GB-BSi5H-6200

Price: ₹36,000



ASRock N3150 Beebox

Price: ₹9,649

RUNNER-UPS

ASRock DeskMini H110 and GIGABYTE BRIX GB-BSi5H-6200

We asked for the best off-the-shelf unit to be sent to us and what came had a Core-i7 6700K, Samsung SM951 and 16 Gigs of RAM. You can see how the DeskMini missed out purely because of the discrete graphics solution. It's also a bit taller than all other units we'd received for the Zero1 Awards and that's because of the socketed processor in the unit.

While the other two contenders were making use of desktop grade components, the Gigabyte BRIX unit on the other hand ends up being the most traditional mini-PC that we received. Though Gigabyte does have mini-PCs with powerful discrete graphics solutions available, they don't sell the same in India. This puts them at a disadvantage against brands that do. The GB-BSi5H-6200 is the most compact of all the units we received which earns them brownie points because the mini-PC segment favours reduced footprint and volume. And while it does lack a discrete graphics solution, it is sufficiently powerful for casual gaming.

BEST BUY

ASRock N3150 Beebox

The Best Buy of our January 2016 benchmark still retains that title for providing a feature set that most other mini-PCs would charge extra for. It's the only one which comes with a USB Type-C port and a multimedia remote in its price segment. And while the Intel N3150 isn't powerful to handle gaming, it handles every other productivity software with ease. A fully decked up unit can cost about ₹ 26,599 which takes away any worries about having to customise the hardware per your needs. **d**